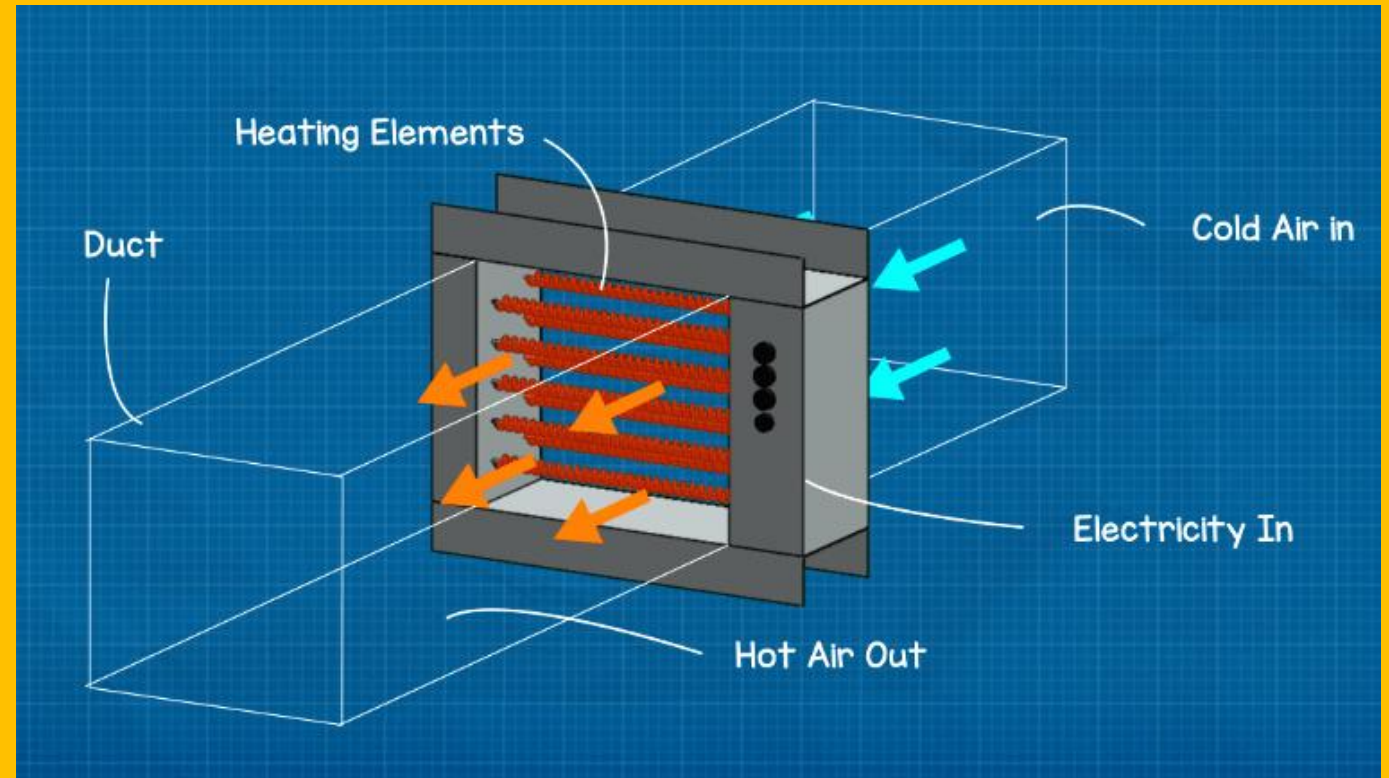


Winner Engineering Learning Hub Series

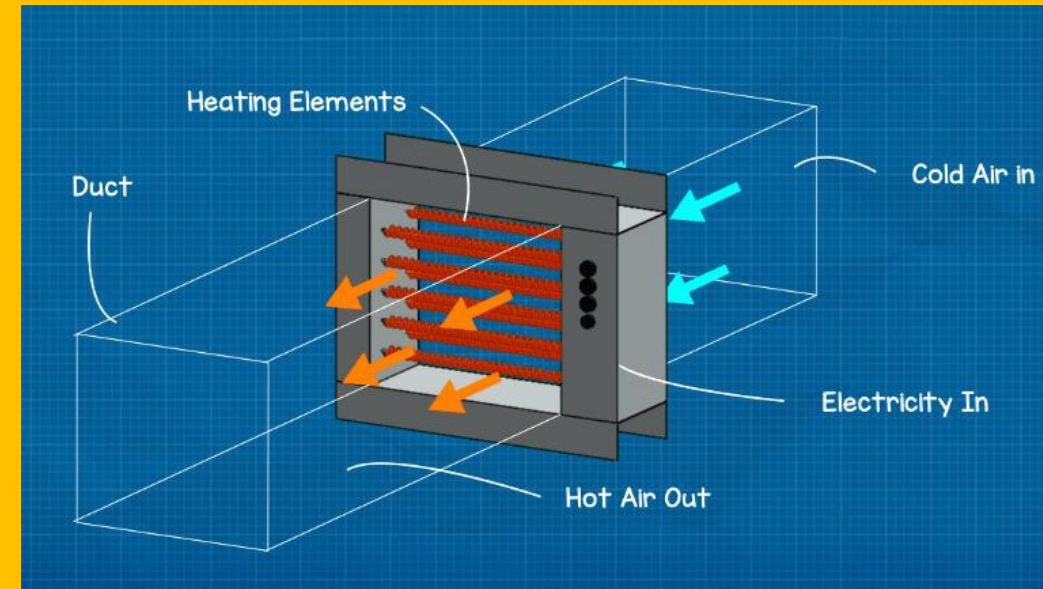
Duct Heater



What is Duct Heater?

A heating device installed in air ducts to provide heat as air passes through the system.

How it Works: To raise the temperature of the air that flows through the duct to control the climate in a building.



When to use Duct Heaters?

Humidity Control: Singapore's high humidity can cause condensation in ductwork. In some cases, duct heaters are used to reheat cooled air to prevent condensation and manage moisture levels, which can help avoid mold growth in air-conditioned spaces.



Data Centres
40-60 RH%
18-27°C



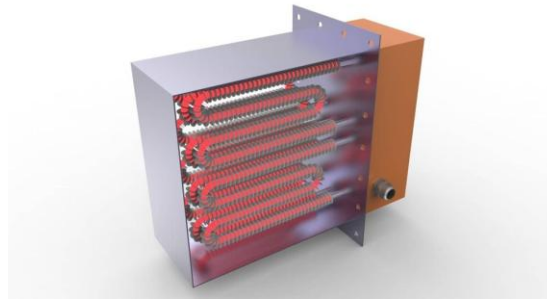
Clean Room
30-60 RH %
18-24 °C



Dry Food Processing and Storage Facilities
40-50 RH%
18-24 °C

Types of Duct Heaters

1. Electric
2. Hot Water
3. Steam
4. Gas Fired



Commonly Used

Comparisons of Duct Heaters

Type	Power Source	Applications	Efficiency	Cost
Electric	Electricity	Offices, Hotel	High	Moderate
Hot Water	Boiler	Factories	Very High	Low
Steam	Steam	Industrial	Moderate	High
Gas-Fired	Natural Gas	Large Building	Moderate	Moderate

How to Select Right Size of Duct Heater?

- **Airflow Rate (CMH):** The volume of air moving through the ducts.
- **Desired Temperature Rise:** The difference between the current and required temperature.
- **Duct Dimensions:** Size of the duct system where the heater will be installed.
- **Heating Load Calculations:** Based on the heat loss of the building.
- **Voltage and Power Supply:** Ensure compatibility with the power system.

$$Heater\ Size(kW) = \frac{L/s * \Delta T}{837}$$

*Formular from Heatrex

Example of Sizing Duct Heater

Space Requirement

12 Air Change: 800CMH

Relative Humidity: 50%

Space Temperature (DB): 24degC

(T.Dew = 12.94degC)

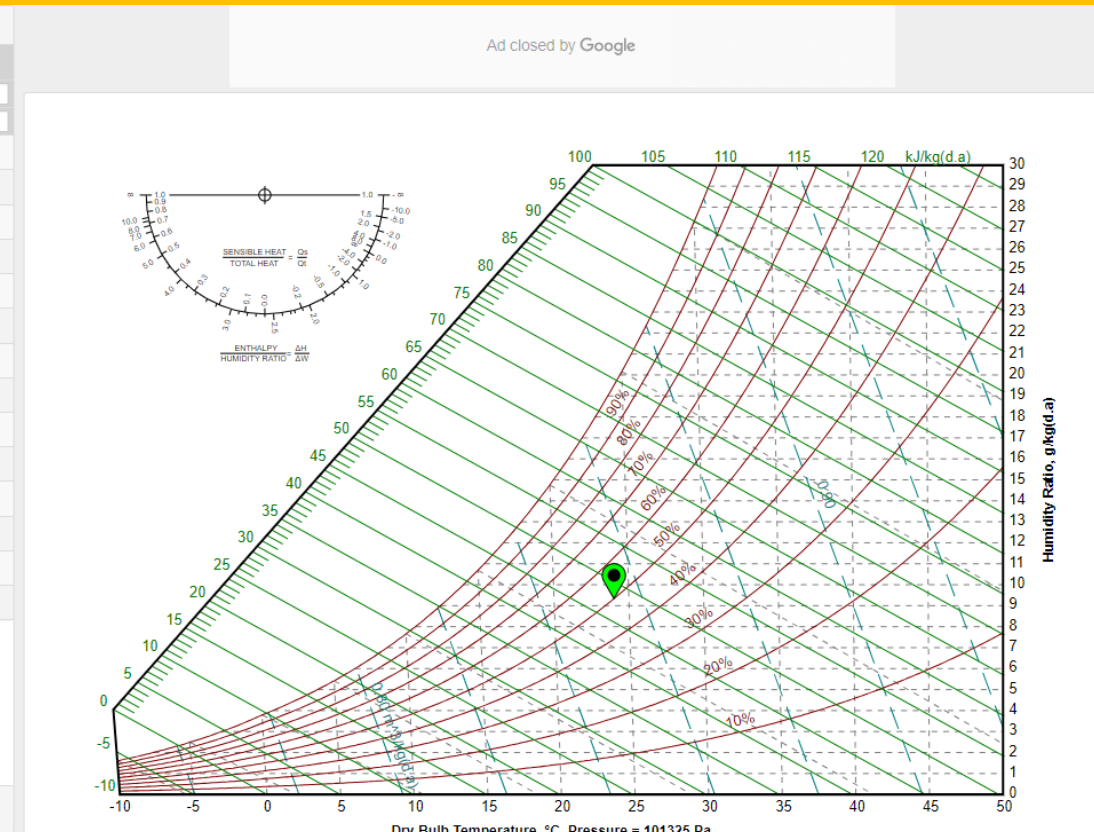
Click on chart for air properties

☒ Or input data for air properties

T.Dry, °C	24
Rel.Humid, %	50

Quantity	Value	Units
P.Ambient	101325	Pa
T.Dry.Bulb	24.000	°C
Humid.Ratio	9.336	g/kg(d.a)
Rel.Humid	50.000	%
T.Wet.Bulb	17.056	°C
T.Dew	12.948	°C
T.Saturation	16.989	°C
Enthalpy	47.895	kJ/kg(d.a)
P.Vapour	1498.453	Pa
P.Sat.Vapour	2985.127	Pa
Spec.Heat	1.020	kJ/(kg.K)
Spec.Volume	0.854	m³/kg(d.a)
Density	1.181	kg/m³

Submit



Example of Sizing Duct Heater

Space Requirement

12 Air Change: 800CMH

Relative Humidity: 50%

Space Temperature (DB): 24degC

(T.Dew = 12.94degC)

Ensure AHU Off-Coil at least **1degC**
lower than T.Dew to meet RH
requirement

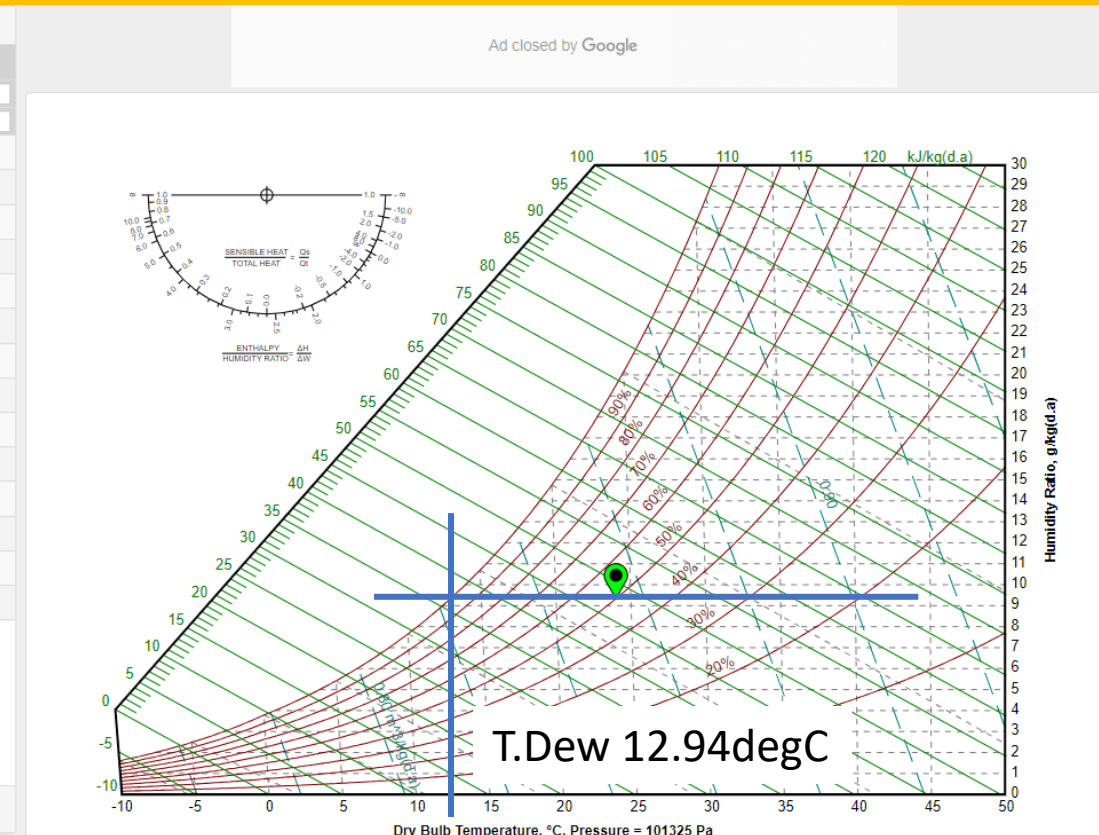
Click on chart for air properties

☒ Or input data for air properties

T.Dry, °C

Rel.Humid, %

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Example of Sizing Duct Heater

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requirement

$$Heater\ Size(kW) = \frac{L/s * \Delta T}{837}$$

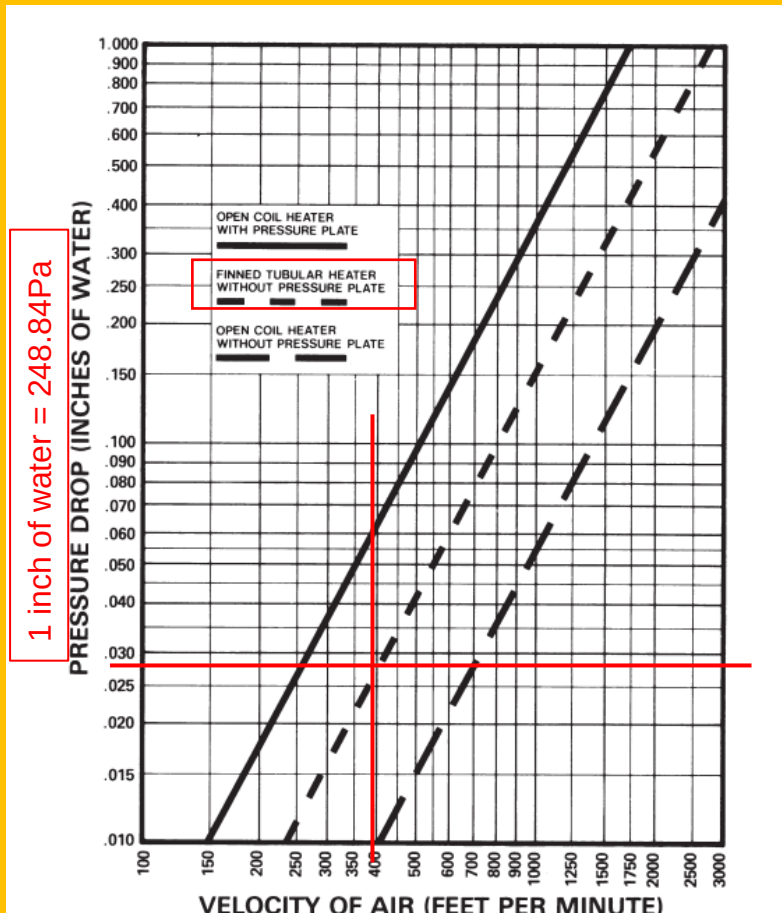
$$Heater\ Size(kW) = \frac{(\frac{800}{3.6}) * (24 - 11.94)}{837}$$

$$Heater\ Size(kW) = 3.2kW$$

*Assumption: 40% of the sensible load are from the space

$$Heater\ Size(kW) = 3.2kW * 60\% = 1.92kW$$

Pressure Drop Calculation



Eg. Air velocity 4m/s = 393FPM

From Chart , pressure drop = 0.27 inches of water

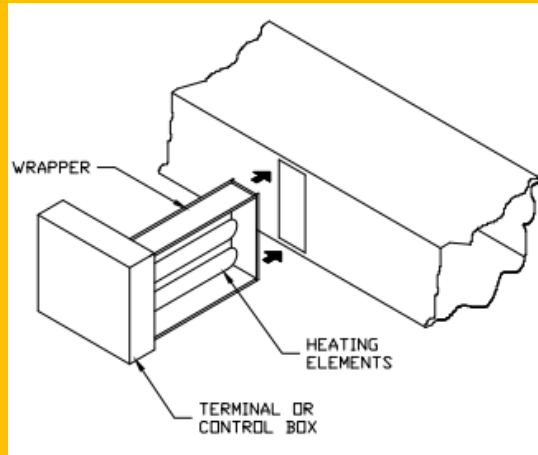
$$0.27 * 248.8\text{pa} = \mathbf{67\text{pa}}$$

1m/s = 196.85FPM

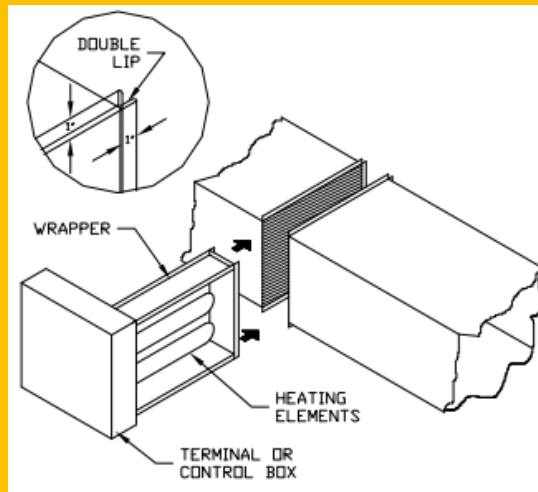
Pressure Drop Chart from Heatrex

Installation Detail

Types Of Installation



Slip-In Heater



Flanged Heater

Aspect	Slip-In Heater	Flanged Heater
Installation	Easier, quicker installation	More complex, requires bolting
Cost	Generally more affordable	More expensive due to additional material and design
Mounting	Less secure, potential shifting or vibration	Secure, rigid mounting with tight seal
Application Size	Best for smaller or lower-demand systems	Best for large, high-pressure, or industrial systems
Sealing	May lead to air leaks	Provides tight, secure seal
Flexibility	Flexible placement, suitable for tight spaces	Less flexible, requires precise positioning
Durability	Less durable for high-vibration or heavy-duty usage	More durable for heavy-duty applications
Maintenance	Easier access due to simpler installation	More difficult to remove or maintain once installed

Installation Detail Requirements

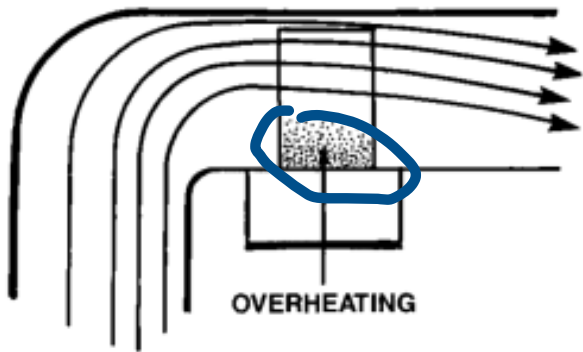
1. Ensure Minimum Airflow is met.
2. Minimum Distance
 - 1m downstream from fan/filters
 - 0.6m straight duct after heater
3. Sensors to install at least 1.5-2m
4. Clearance from combustible material at least 150mm
5. Avoid placing heater with Internal Insulation
6. Access for Maintenance
7. Provision of Power for Duct Heater
8. Interlock Duct Heater with Equipment

KW	STANDARD VOLTAGES					DIMENSIONS IN. (MM)		MINIMUM AIR FLOW CFM (m ³ /MIN)	NO. OF ELEMS.	CATALOG FLANGE TYPE / QFD	NUMBERS INSERT TYPE / QID	APPROX. WEIGHT LBS (KG)
	120V 1φ	208V 1φ	240V 3φ	480V 1φ	600V 3φ	A	B					
✓ 1	✓	✓	—	—	—	6 (152)	6 (152)	✓ 50 (1.4)	2	QFD0010	QID0010	15 (7)
2.5	✓	✓	✓	—	—	6 (152)	6 (152)	150 (4.2)	3	QFD0025	QID0025	15 (7)
7.5	—	✓	✓	✓	✓	14 (356)	8 (203)	390 (11.0)	4	QFD0075	QID0075	20 (9)
✓ 10	—	✓	✓	✓	✓	14 (356)	12 (305)	✓ 500 (14.1)	6	QFD0100	QID0100	25 (11)
12.5	—	✓	✓	✓	✓	16 (406)	12 (305)	625 (17.7)	6	QFD0125	QID0125	30 (14)
15	—	✓	✓	✓	✓	18 (457)	12 (305)	750 (21.2)	6	QFD0150	QID0150	30 (14)
17.5	—	✓	✓	✓	✓	18 (457)	14 (356)	875 (24.8)	6	QFD0175	QID0175	35 (16)
20	—	✓	✓	✓	✓	18 (457)	16 (406)	1000 (28.3)	6	QFD0200	QID0200	35 (16)
25	—	—	✓	✓	✓	20 (508)	18 (457)	1250 (35.4)	9	QFD0250	QID0250	50 (23)
30	—	—	✓	✓	✓	24 (610)	18 (457)	1500 (42.4)	9	QFD0300	QID0300	55 (25)
35	—	—	✓	✓	✓	24 (610)	22 (559)	1650 (46.7)	9	QFD0350	QID0350	60 (27)
40	—	—	✓	✓	✓	26 (660)	24 (610)	2050 (58.0)	12	QFD0400	QID0400	70 (32)
45	—	—	✓	✓	✓	28 (711)	24 (610)	2200 (62.2)	12	QFD0450	QID0450	75 (34)
50	—	—	✓	✓	✓	28 (711)	26 (660)	2500 (70.7)	12	QFD0500	QID0500	80 (36)
60	—	—	✓	✓	✓	30 (762)	30 (762)	3000 (84.9)	15	QFD0600	QID0600	95 (43)
80	—	—	✓	✓	✓	36 (914)	32 (813)	4000 (113.1)	15	QFD0800	QID0800	105 (48)
100	—	—	✓	✓	✓	42 (1067)	36 (914)	5250 (148.5)	18	QFD1000	QID1000	130 (59)
120	—	—	✓	✓	✓	48 (1219)	36 (914)	6000 (170.0)	18	QFD1200	QID1200	150 (68)

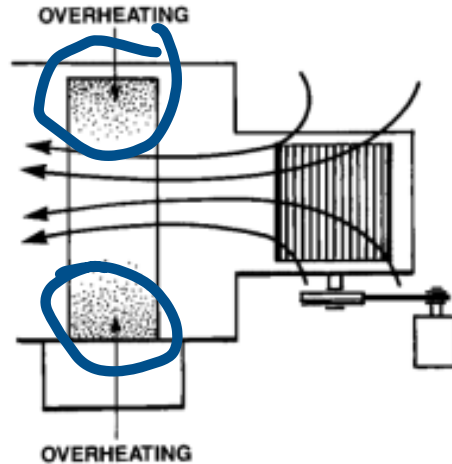
Example of difference duct heater sizes and minimum airflow

Installation Detail

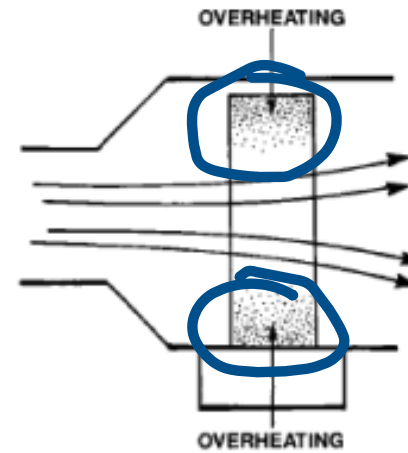
Heater Misapplication



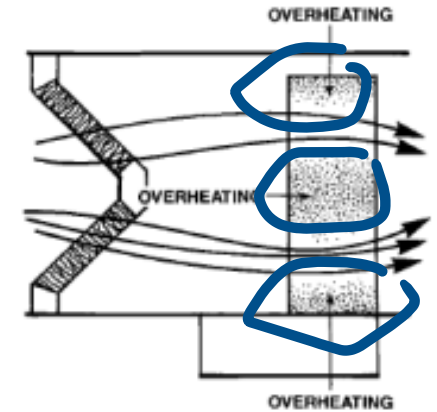
Heater too close to elbow



Heater adjacent to transition



Heater too close to fan



Heater partially blocked by filter or frame member

Conclusion

Ensure the heater is properly sized for the duct and airflow.

- Install it downstream of filters, and not too close to other components like fans or air conditioning coils.
 - Provide proper access for maintenance and cleaning.
 - Make sure there's adequate space for wiring and electrical connections.
- Consider clearances from combustible materials, especially for gas-fired heaters.

Safety Considerations: Use overheat protection devices and airflow sensors.